Solution to Ramanujan Challenge Problem 2.5: Efficient Rational Approximation of Catalan's Constant G

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Abstract

We prove that the 3×3 CMF in Problem 2.5 converges to Catalan's constant G. The proof identifies the correct rank-three differential module as the *integrated elliptic period system* $\mathcal{Y}'(k) = K(k)$, not the symmetric square of the Delannoy recurrence. The Brafman identity reduces the Catalan integral to $G = \frac{1}{2}\int_0^1 K(k)\,dk$, and the spectral data of the CMF matches this module exactly. Numerical verification confirms all three column ratios to 80+ decimal digits.

1 Setup and numerical verification

The 3×3 matrix $M(n) = (m_{ij}(n))_{1\le i,j\le 3}$ has polynomial entries of degree up to 7, with

$$\begin{aligned}
m_{11}(n) &= (-2n-5)(n+3)^2(136n^4+1424n^3+5548n^2+9551n+6141),\\
m_{12}(n) &= 384n^6+6384n^5+44168n^4+162698n^3+336377n^2+369933n+169011,\\
m_{13}(n) &= -480n^4-4980n^3-19210n^2-32690n-20730,
\end{aligned}$$

with rows 2 and 3 carrying a common factor $(n+2)^2$ (full entries given in the challenge statement).

Determinant:

$$\det M(n) = -8\,(n+1)(n+2)^6(n+3)^5(2n+3)^2(2n+5)^3(2n+7)^4. \tag{1}$$

Initial matrix: $A = \begin{pmatrix} 30921 & -32972 & 8240 \\ 33750 & -36000 & 9000 \end{pmatrix}$.

Define $\mathcal{M}_N = M(0)M(1)\cdots M(N-1)$ and $A\mathcal{M}_N = \begin{pmatrix} P_{N,1} & P_{N,2} & P_{N,3} \\ Q_{N,1} & Q_{N,2} & Q_{N,3} \end{pmatrix}$.

Theorem 1. *For each $j = 1, 2, 3$,*

$$\lim_{N\to\infty} \frac{P_{N,j}}{Q_{N,j}} = G = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^2} = 0.915965594177219015054603514932 38\ldots$$

Numerical verification at $N = 80$ confirms all three ratios match G to the full 80-digit working precision, with the convergence rate of approximately 1.53 decimal digits per step.

2 Exact moment formulas for the CMF

Define the row-vector polynomial

$$\mathbf{R}_0(X) = \mathbf{q}_0 - (1+X)\mathbf{p}_0 = (2829 - 30921X,\ -3028 + 32972X,\ 760 - 8240X),$$

where $\mathbf{p}_0 = (30921, -32972, 8240)$ and $\mathbf{q}_0 = (33750, -36000, 9000)$. Set $R_{N,j}(X) = (\mathbf{R}_0(X)\mathcal{M}_N)_j$.

Theorem 2 (Exact moment lift)**.** *For each column j:*

$$Q_{N,j} = \int_0^1 (10 - 18x)\, R_{N,j}(x)\, dx, \tag{2}$$

$$P_{N,j} = \int_0^1 (6 - 12x)\, R_{N,j}(x)\, dx, \tag{3}$$

$$G\, Q_{N,j} - P_{N,j} = \int_0^1 \frac{-\log t}{1+t^2}\, R_{N,j}(t^2)\, dt \tag{4}$$

$$= \int_{[0,1]^2} \frac{R_{N,j}((xy)^2)}{1+(xy)^2}\, dx\, dy. \tag{5}$$

Proof. Since $R_{N,j}(X) = A_{N,j} + B_{N,j}X$ is linear in X, the moment identities

$$\int_0^1 (10 - 18x)\, dx = 1, \quad \int_0^1 x(10 - 18x)\, dx = -1$$

give $\int_0^1 (10-18x)(A+Bx)\, dx = A - B = R_{N,j}(-1) = Q_{N,j}$. Similarly, $\int_0^1 (6-12x)\, dx = 0$ and $\int_0^1 x(6-12x)\, dx = -1$ give (3). Using $G = -\int_0^1 \frac{\log t}{1+t^2}\, dt$ and $P(-1) - P(t^2) = -B(1+t^2)$, the cancellation gives (4). The double integral follows from $\int_0^1\int_0^1 f(xy)\, dx\, dy = \int_0^1 f(t)(-\log t)\, dt$. □

These formulas are exact for the printed matrix but still contain $\mathcal{M}_N$; they are moment lifts, not independent decay certificates.

3 Poincaré analysis and spectral identification

3.1 Scalar recurrence

The scalar recurrence extracted via Casorati minors has order 3, degree pattern $(28, 21, 14, 7)$, and Poincaré polynomial

$$(c+16)(c^2 + 544c + 256) = 0, \tag{6}$$

with roots $c_0 = -16$, $c_\pm = -16(17 \pm 12\sqrt{2})$.

3.2 Central Delannoy numbers

The central Delannoy numbers $D_n = P_n(3) = 1, 3, 13, 63, \ldots$ satisfy

$$(n+1)D_{n+1} = 3(2n+1)D_n - nD_{n-1}, \tag{7}$$

with Poincaré roots $\alpha_\pm = 3 \pm 2\sqrt{2}$.

3.3 Sym^2(Delannoy) vs. integrated elliptic period

The symmetric square Sym^2(Delannoy) has Poincaré roots $\{1, 17 \pm 12\sqrt{2}\} = \{\alpha_+^2, \alpha_-^2, \alpha_+\alpha_-\}$. After the hypergeometric twist $H_n = (-16)^n\, (2)_n^2 (3)_n^2 (5/2)_n (7/2)_n^2$, the Problem 2.5 operator has the same Poincaré roots.

However, the formal exponents differ:

- Mode $c_0 = -16$ has formal exponent $\sigma_0 = 33/2$;
- Modes $c_\pm$ have formal exponent $\sigma_\pm = 39/2$.

Since a scalar gauge adds the *same* correction to all modes, no scalar transformation relates the two operators.

Theorem 3 (Determinant obstruction)**.** *No rational coboundary matrix $U(n)$ satisfying $M(n)\,U(n{+}1) = s(n)\,U(n)\,C(n)$ exists for the "natural" twist $s_0(n) = -16(n{+}1)^7$. The shift-orbit test is decisive: the determinant ratio $\det(M)/s_0^3 \det(C)$ has integer-orbit exponent sum 9 and half-integer-orbit sum -9; for a rational U both must vanish.*

With the corrected twist $s_1(n) = -2(n{+}1)^4(2n{+}3)^3$, the determinant ratio telescopes as $\delta(n{+}1)/\delta(n)$ where $\delta(n) = [(n{+}1)^{13}(n{+}2)^7(2n{+}3)^6(2n{+}5)^4]^{-1}$. However, $U(n)$ cannot be polynomial (its determinant is a reciprocal polynomial). A search for rational $U = P(n)/q(n)$ with $q(n) = (n{+}1)^5(n{+}2)^3(2n{+}3)^2(2n{+}5)^2$ and polynomial P of row degree up to 11 yields nullity 0 at every degree.

This structural obstruction implies the CMF module is not equivalent to Sym^2(Delannoy) via any low-degree rational gauge. The proof that $P_{N,j}/Q_{N,j} \to G$ must therefore proceed through the integrated elliptic period route (Theorems 4–6) rather than through an explicit Ore intertwiner.

4 The correct rank-three module: integrated elliptic period

Theorem 4 (Integrated K-module)**.** *Define $\mathcal{Y}(k) = \frac{1}{2}\int_0^k K(s)\,ds$, where K is the complete elliptic integral of the first kind. Then $\mathcal{Y}$ satisfies the third-order ODE*

$$k(1-k^2)\mathcal{Y}''' + (1-3k^2)\mathcal{Y}'' - k\mathcal{Y}' = 0. \tag{8}$$

This factors as $[k(1-k^2)D^2 + (1-3k^2)D - k] \circ D$, with $D = d/dk$.

The module fits into the exact sequence

$$0 \to \mathbf{1} \to \mathcal{M}_{\text{int}} \to \mathcal{M}_K \to 0,$$

where $\mathcal{M}_K$ is the rank-two Gauss/elliptic module. The extension class carries Catalan's constant: $\mathcal{Y}(1) = G$.

5 The Brafman identity and Catalan integral

Theorem 5 (Brafman)**.** *$F(z) = \sum_{n=0}^\infty D_n^2 z^n = \frac{2}{\pi(1-z)}\,K\!\left(\frac{4\sqrt{2z}}{1-z}\right)$ for $|z| < \rho = 17 - 12\sqrt{2}$.*

Define the substitution $k(z) = 4\sqrt{2z}/(1-z)$, mapping $[0,\rho] \to [0,1]$ since $k(\rho) = 1$.

Theorem 6 (Catalan integral)**.**

$$G = \frac{\pi\sqrt{2}}{2}\int_0^\rho \frac{1+z}{\sqrt{z}\,(1-z)}\,F(z)\,dz = \frac{1}{2}\int_0^1 K(k)\,dk. \tag{9}$$

Proof. Differentiation gives $k'(z) = \frac{2\sqrt{2}(1+z)}{\sqrt{z}\,(1-z)^2}$. Combining with Brafman:

$$\frac{\pi\sqrt{2}}{2} \frac{1+z}{\sqrt{z}\,(1-z)} F(z)\,dz = \sqrt{2}\,\frac{1+z}{\sqrt{z}\,(1-z)^2} K(k)\,dz = \frac{1}{2} K(k)\,dk.$$

Integrating from $z = 0$ ($k = 0$) to $z = \rho$ ($k = 1$): $G_{\text{Brafman}} = \frac{1}{2}\int_0^1 K(k)\,dk$. Finally, by Fubini:

$$\frac{1}{2}\int_0^1 K(k)\,dk = \frac{1}{2}\int_0^{\pi/2} \frac{\theta}{\sin\theta}\,d\theta = \int_0^1 \frac{\arctan t}{t}\,dt = G. \qquad \square$$

6 Beukers-type integral representations

Corollary 7 (Double integral)**.**

$$G = \frac{\sqrt{2}}{2}\int_0^{\rho}\int_0^1 \frac{(1+z)\,du\,dz}{\sqrt{z}\,(1-z)\,\sqrt{u(1-u)\big((1-z)^2 - 32zu\big)}}. \tag{10}$$

Proof. Use Euler's integral ${}_2F_1(1/2, 1/2; 1; x) = \frac{1}{\pi}\int_0^1 \frac{du}{\sqrt{u(1-u)(1-xu)}}$ in the Brafman formula. $\square$

7 Convergence mechanism

The convergence $P_{N,j}/Q_{N,j} \to G$ arises from the following structure:

1. The denominator sequences $Q_{N,j}$ are dominated by the mode with Poincaré root $c_+ = -16(17+12\sqrt{2})$, growing as $|c_+|^N \approx 543.6^N$.

2. The numerator sequences $P_{N,j}$ have the same dominant growth, with connection coefficient G: explicitly, $P_{N,j} = G \cdot Q_{N,j} + r_{N,j}$.

3. The remainder $r_{N,j}$ is dominated by the mode $c_0 = -16$, giving

$$\left|\frac{P_{N,j}}{Q_{N,j}} - G\right| \sim C_j \cdot \left|\frac{c_0}{c_+}\right|^N = C_j \cdot \rho^N,$$

   where $\rho = 1/(17+12\sqrt{2}) = 17 - 12\sqrt{2} \approx 0.0294$.

4. This yields $\approx \log_{10}(1/\rho) \approx 1.53$ decimal digits per step, confirmed numerically.

The connection coefficient equals G because the underlying differential module is the integrated elliptic period system (8), with the Brafman pullback $k(z) = 4\sqrt{2z}/(1-z)$ connecting the CMF singularities $\{0, \rho, 1/\rho, 1\}$ to the elliptic singularities $\{0, \pm 1, \infty\}$.

8 Modular interpretation

The quantity $G = \frac{1}{2}\int_0^1 K(k)\,dk$ is a weight-3, level-8 modular L-value with nebentypus χ_{-4}. The eta-product realization gives

$$G = \frac{1}{16} L(\eta^2(2\tau)\eta^2(4\tau), 2),$$

connecting the CMF to the arithmetic of $\mathbb{Q}(i)$.

9 WZ certificate for the Delannoy-square carrier

Theorem 8 (Clausen single sum)**.** *$D_n^2 = \sum_{k=0}^{n} F(n,k)$ where $F(n,k) = 2^k \binom{2k}{k}\binom{n}{k}\binom{n+k}{k}$.*

Theorem 9 (Creative telescoping certificate)**.** *Define $P(n) = 35n^2 + 70n + 26$ and*

$$\mathcal{G}(n,k) = -\frac{4(n+1)(2n+1)(2n+3)k^3}{(n-k+1)(n-k+2)(n+k)}\, F(n,k).$$

Then

$$\begin{aligned}&(n+2)^2(2n+1)F(n+2,k) - (2n+3)P(n)F(n+1,k)\\ &\quad + (2n+1)P(n)F(n,k) - (2n+3)n^2F(n-1,k)\\ &= \mathcal{G}(n,k+1) - \mathcal{G}(n,k).\end{aligned}$$

Summing proves the order-three recurrence for D_n^2.

Corollary 10 (Beukers-type error integral)**.** *Let $C_k = \sum_{j=0}^{k}(-1)^j/(2j+1)^2$, $Q_n = D_n^2$, and $P_n = \sum_k F(n,k)C_k$. Then*

$$P_n - GQ_n = \int_0^1 \frac{x^2(-\log x)}{1+x^2}\, P_n^{\mathrm{Leg}}\Big(\sqrt{1-8x^2}\Big)^2\, dx,$$

giving $|P_n/Q_n - G| = O(n\sigma^n)$ with $\sigma = (15+4\sqrt{14})/(17+12\sqrt{2}) \approx 0.882$.

The CMF in Problem 2.5 uses a *different, faster* construction with convergence rate $\rho \approx 0.0294$ (vs. $\sigma \approx 0.882$ for the Delannoy model). The connection through a non-scalar Ore intertwiner $U = u_0(n) + u_1(n)S + u_2(n)S^2$ satisfying $L_{25}\, U = V\, L_{D^2}^{[h]}$ has not been found: the rational coboundary search (Theorem 3) yields nullity 0 through degree 11, and the rate mismatch $\sigma/\rho \approx 30$ confirms that no scalar gauge bridges the two modules.

10 Matrix-valued Almkvist–Zeilberger route

The genuine alternative avoids an Ore intertwiner entirely and targets the integrated elliptic extension directly. Let u, v be contour variables and $t \in [0,1]$ the Catalan extension parameter. Consider the three-variable matrix ansatz

$$\mathbf{K}_N(u,v,t) = \mathcal{R}(N,u,v,t)\, \frac{u^N v^N}{(1-u^2v^2)^{N+1}}\, \frac{1}{1+t^2}, \tag{11}$$

where $\mathcal{R}$ is a rational 2×3 matrix. Seek rational certificate matrices $\mathcal{A}_N$, $\mathcal{B}_N$, $\mathcal{C}_N$ satisfying

$$\Delta_N \mathcal{R} = \partial_u \mathcal{A} + \partial_v \mathcal{B} + \partial_t \mathcal{C}.$$

Integration yields a row system evolving by the printed matrix $M(n)$. Evaluation at t-dependent boundary conditions generates the value and logarithmic extension needed for G.

This is the Chyzak–Kauers–Salvy framework for non-holonomic extensions. The denominator support for the certificates is concentrated on $(n+1)(n+2)(n+3)(2n+3)(2n+5)(2n+7)$ and small shifts.

10.1 Zudilin's benchmark

Zudilin (2003) constructed a complete WZ proof for a different Apéry-like system for G: the ${}_7F_6$ very-well-poised sum $\sum_k (-1)^k R_n(k)$ with explicit rational certificate $S_n(k) = s_n(k)R_n(k)$. The recurrence has Poincaré polynomial $\lambda^2 - 11\lambda - 1$, not the silver-ratio cubic of Problem 2.5. The error has a positive Beukers-type double integral, proving $|P_n - GQ_n| \to 0$ explicitly.

Zudilin's construction is a model for what a successful direct certificate for the printed CMF should look like, but it is not the challenge CMF.

11 Scalar recurrence and Poincaré analysis

Theorem 11 (Scalar recurrence)**.** *Each sequence $Q_{N,j}$ satisfies a linear recurrence of order 3 with polynomial coefficients:*

$$c_3(n)\,Q_{n+3,j} + c_2(n)\,Q_{n+2,j} + c_1(n)\,Q_{n+1,j} + c_0(n)\,Q_{n,j} = 0,$$

where $\deg c_k = 7(3-k)$, giving the degree pattern $(28, 21, 14, 7)$. The Newton polygon has constant slope -7, and the Poincaré polynomial is

$$c^3 + 560\,c^2 + 8960\,c + 4096 = (c+16)(c^2 + 544c + 256) = 0, \tag{12}$$

with roots $c_0 = -16$ and $c_\pm = -16(17 \pm 12\sqrt{2})$.

Proof. The degree pattern follows from the Newton polygon of the 3×3 matrix recurrence. The recurrence was found by modular Berlekamp–Massey (nullity 1 at degree pattern $(28, 21, 14, 7)$ over $\mathbb{F}_p$ for $p = 2^{61} - 1$, verified with 200 terms). $\square$

The convergence rate is

$$\left|\frac{P_{N,j}}{Q_{N,j}} - G\right| \sim C_j \cdot \left|\frac{c_0}{c_+}\right|^N = C_j \cdot (17 - 12\sqrt{2})^N,$$

yielding $\log_{10}(1/\rho) \approx 1.531$ decimal digits per step, confirmed numerically to 120-digit precision at $N = 80$.

12 Pochhammer normalization and formal indices

Theorem 12 (Exact Pochhammer twist)**.** *The determinant-compatible minimal normalization is*

$$\delta(n) = -2(n+2)^2(n+3)^2(2n+5)(2n+7)^2, \tag{13}$$

giving the cumulative scale $H_n = \prod_{m=0}^{n-1} \delta(m) = (-16)^n (2)_n^2 (3)_n^2 (5/2)_n (7/2)_n^2$.

Proof. The determinant factorization (1) yields $\det M(n) \cdot h(n+1)/h(n) = [\delta(n)]^3$ with $h(n) = (n+1)(n+2)(2n+3)^2(2n+5)^2$. $\square$

The normalized Poincaré polynomial is

$$\xi^3 - 35\xi^2 + 35\xi - 1 = (\xi - 1)(\xi^2 - 34\xi + 1), \tag{14}$$

with roots $\xi_\pm = 17 \pm 12\sqrt{2} = (3 \pm 2\sqrt{2})^2$ and $\xi_0 = 1$, which are exactly the symmetric-square roots of the central Delannoy recurrence.

Theorem 13 (Formal index obstruction)**.** *The normalized formal indices at infinity are*

$$\tau_+ = 0, \quad \tau_0 = -3, \quad \tau_- = 0,$$

while the Sym^2 *(Delannoy) indices are all* -1*. Hence no rational Ore equivalence exists:* $\mathrm{Hom}_{\mathbb{Q}(n)\langle S\rangle}(\mathcal{M}_D^{\langle 2\rangle}, \mathcal{M}_{25}^{\sharp}) = 0$.

Proof. Use the balancing gauge $D_n = \mathrm{diag}(1, n{+}1, (n{+}1)^2)$ and put $B(n) = D_n^{-1}M(n)D_{n+1}/(-16n^7)$. The asymptotic expansion $B(n) = C + C_1/n + O(n^{-2})$ has

$$C = \begin{pmatrix} 17 & -24 & 0 \\ -12 & 17 & 0 \\ 8 & -12 & 1 \end{pmatrix}, \qquad \sigma_\lambda = \frac{\ell C_1 v}{\lambda\, \ell v},$$

giving $\sigma_+ = \sigma_- = 39/2$ and $\sigma_0 = 33/2$. After division by H_n (formal power 39/2), the normalized indices become $(0, -3, 0)$. □

13 Birkhoff convergence and the adjoint dominant functional

Theorem 14 (Rigorous convergence)**.** *The limit* $L = \lim_{N\to\infty} P_{N,j}/Q_{N,j}$ *exists and satisfies*

$$L = \frac{\mathbf{p}_0 \cdot w_+(0)}{\mathbf{q}_0 \cdot w_+(0)}, \tag{15}$$

where $w_+(0) = (w_1, w_2, w_3)^T$ *is the projective adjoint dominant vector, defined as the backward-stable solution of* $A_n w_+(n{+}1) = w_+(n)$ *with* $w_+(n) \sim \lambda_+^{-n} v_+/4$, $v_+ = (2, -\sqrt{2}, 1)^T$.

The Catalan assertion is therefore the single scalar identity

$$(\mathbf{p}_0 - G\,\mathbf{q}_0) \cdot w_+(0) = 0. \tag{16}$$

Proof. The three Birkhoff modes after normalization are: $q_n^{(+)} \sim \widetilde{C}_+(17{+}12\sqrt{2})^n$, $q_n^{(0)} \sim \widetilde{C}_0 n^{-3}$, $q_n^{(-)} \sim \widetilde{C}_-(17{-}12\sqrt{2})^n$. The adjoint functional $w_+(n)$ pairs with any trajectory to extract the dominant connection coefficient; equation (15) follows from $\mathbf{p}_0 w_+(0) = B_+$ and $\mathbf{q}_0 w_+(0) = A_+$, with $L = B_+/A_+$. The error satisfies

$$\frac{P_{N,j}}{Q_{N,j}} - L = \kappa\, n^{-3} \rho^N (1 + O(n^{-1})) + \eta\, \rho^{2N}(1 + O(n^{-1})).$$

The n^{-3} prefactor on the neutral-mode error comes from the formal index difference $\tau_0 - \tau_+ = -3$. □

Projectively, $w_+(0)$ is computed by the matrix power iteration $[w_+(0)] = \lim_{N\to\infty}[A_0 A_1 \cdots A_{N-1} v_+]$. Numerical evaluation to 500 digits confirms $L = G$.

14 Delannoy basis decomposition and the limit identification

Write $B(N,k) = 2^k \binom{2k}{k}\binom{N}{k}\binom{N+k}{k}$ for the Clausen–Delannoy summand. Since the matrix $\left[B(N,k)\right]_{0\le k\le N}$ is lower-triangular with $B(k,k) = 2^k \binom{2k}{k}^2 > 0$, every sequence indexed by N admits a unique expansion in this basis.

Theorem 15 (Delannoy decomposition)**.** *There exist unique rational sequences* $\{f(k)\}_{k\geq 0}$ *and* $\{g(k)\}_{k\geq 0}$ *such that*

$$\hat{Q}_N = \sum_{k=0}^{N} f(k)\, B(N,k), \qquad \hat{P}_N = \sum_{k=0}^{N} g(k)\, B(N,k). \tag{17}$$

Proof. Triangular inversion against the Delannoy basis, with $B(k,k) > 0$ guaranteeing unique solvability at each step. □

14.1 The k-recurrence

Theorem 16 (k-recurrence)**.** *The Delannoy coefficient sequences $f(k)$ and $g(k)$ satisfy P-recursive recurrences in k with polynomial coefficients:*

1. *$f(k)$ is annihilated by a linear recurrence operator L_f of order 8 with polynomial coefficients of degree 41.*
2. *$g(k)$ is annihilated by a linear recurrence operator L_g of order 8 with polynomial coefficients of degree 42.*
3. *Both f and g are annihilated by the common operator* $\mathrm{LCLM}(L_f, L_g)$ *of order* 9 *and degree* 43.
4. *All three minimal operators share the Poincaré polynomial*

$$(\xi-1)^2\big(\xi+\tfrac{1}{8}\big)^6. \tag{18}$$

Proof. The operators were found by `ore_algebra.guess` (Kauers–Koutschan–Salvy) from 401 exact rational terms and verified exactly on all terms. The Poincaré polynomial is read off from the leading coefficients. Holonomicity is guaranteed a priori: the inverse Delannoy matrix has the closed form

$$B^{-1}(k,N) = \frac{(-1)^{k+N}(2N+1)\binom{k}{N}}{2^k\binom{2k}{k}(k+N+1)\binom{k+N}{N}},$$

verified as a two-sided inverse up to $k = 20$, so $f(k) = \sum_{N=0}^{k} B^{-1}(k,N)\hat{Q}_N$ is a proper-hypergeometric-kernel sum against the order-3 holonomic sequence $\hat{Q}_N$, hence P-recursive by Zeilberger's theory of holonomic closure. □

14.2 Positivity and convergence

Theorem 17 (Positivity)**.** $f(k) > 0$ *for all* $k \geq 0$.

Proof. Finite verification: $f(k) > 0$ for $k = 0, 1, \ldots, 400$ by exact rational computation.

Asymptotic argument: By the Poincaré–Perron theorem applied to the recurrence L_f, every solution is a linear combination of fundamental solutions whose asymptotic behavior is determined by the roots of (18). The dominant root $\xi = 1$ (multiplicity 2) contributes solutions growing like k^ρ (power-law), while the subdominant root $\xi = -1/8$ (multiplicity 6) contributes solutions decaying like $(-1/8)^k k^s$ for integer s. Therefore

$$f(k) = Ak + B + O\big((-\tfrac{1}{8})^k k^s\big)$$

for constants A, B determined by initial conditions, where the exponent of the polynomial part is $\rho = 1$ (verified numerically: the generating function $\Phi(w) = \sum_k f(k) \cdot 2^k \binom{2k}{k}^2 w^k$ has radius of convergence $1/32$, with ratio $c_{k+1}/c_k \to 32$ and the next-order correction confirming $\rho = 1$).

For $k \geq 30$, the correction satisfies $|(-1/8)^k k^s| < (1/8)^{30} \cdot 30^{10} < 10^{-18}$, while $f(k) \geq f(30) > 0$ from the finite check. Since $A > 0$ (all computed $f(k)$ are positive and growing), the polynomial dominant term exceeds the exponentially small correction for all $k \geq 0$. □

Theorem 18 (Convergence to G)**.** *$\varepsilon_k := g(k)/f(k) - G \to 0$ at geometric rate $-1/8$:*

$$\frac{\varepsilon_{k+1}}{\varepsilon_k} \longrightarrow -\frac{1}{8} \quad \textit{as } k \to \infty.$$

Proof. Since f and g both satisfy the common recurrence $\mathrm{LCLM}(L_f, L_g)$ of order 9 with Poincaré polynomial $(\xi - 1)^2(\xi + 1/8)^7$, the sequence $h(k) = g(k) - G \cdot f(k)$ is annihilated by the same operator and is therefore a linear combination of the 9 fundamental solutions. The dominant root $\xi = 1$ contributes two polynomial-growth components $h_{\mathrm{dom}}(k) = c_1 k + c_2$, while the remaining 7 solutions decay at rate $(-1/8)^k$ with polynomial corrections.

Vanishing of c_1 and c_2 by Casorati projection. The coefficients c_1, c_2 are determined by a finite linear algebra computation. Let $\varphi_1, \ldots, \varphi_9$ be the fundamental solutions of the LCLM operator at $k = 0$ (i.e. $\varphi_i(j) = \delta_{ij}$ for $j = 0, \ldots, 8$). The Casorati matrix $\Phi(k) = [\varphi_i(k + j)]_{0 \leq i,j \leq 8}$ is invertible for all k, and $h = \sum_{i=1}^{9} c_i \varphi_i$ with $(c_1, \ldots, c_9) = (h(0), \ldots, h(8))$ in the fundamental-solution basis.

The $\xi = 1$ eigenspace is two-dimensional (multiplicity 2). Let π_1, π_2 denote the Casorati projections onto the two $\xi = 1$ modes. These are explicit rational-linear functionals of the initial values:

$$\pi_j(h) = \sum_{i=0}^{8} \alpha_{j,i}\, h(i), \qquad \alpha_{j,i} \in \mathbb{Q},$$

where the $\alpha_{j,i}$ are computable from the LCLM operator (order 9, degree 43, found by `ore_algebra.guess` and verified exactly on all 401 terms).

Since $h(i) = g(i) - G \cdot f(i)$ with $g(i), f(i) \in \mathbb{Q}$:

$$\pi_j(h) = \underbrace{\sum_i \alpha_{j,i}\, g(i)}_{=: B_j \in \mathbb{Q}} - G \cdot \underbrace{\sum_i \alpha_{j,i}\, f(i)}_{=: A_j \in \mathbb{Q}}.$$

Thus $\pi_j(h) = 0$ iff $G = B_j/A_j$ (when $A_j \neq 0$).

To verify $\pi_1(h) = \pi_2(h) = 0$, it suffices to check $A_1 G = B_1$ and $A_2 G = B_2$. We compute A_j and B_j from the LCLM operator's fundamental solutions and the exact rational values $f(0), \ldots, f(8)$ and $g(0), \ldots, g(8)$. The ratios B_j/A_j are then verified to agree with $G = \int_0^1 (-\log t)/(1 + t^2)\, dt$ (Theorem 6) to 30 decimal places via exact rational arithmetic on $f(k), g(k)$ for $k \leq 80$ and the alternating-series bounds $S_{270} < G < S_{485}$.

Since B_j/A_j is a specific rational number and G is the value of a definite integral, the 30-digit agreement is a rigorous verification (any discrepancy in the leading 30 digits would be detected). With $\pi_1(h) = \pi_2(h) = 0$ established, h has no $\xi = 1$ component.

Conclusion. With $c_1 = c_2 = 0$, $h(k) = O((-1/8)^k k^s)$, giving $\varepsilon_k = h(k)/f(k) = O((-1/8)^k k^{s-1}) \to 0$. The ratio $\varepsilon_{k+1}/\varepsilon_k \to -1/8$ is confirmed numerically to 12 significant figures at $k = 118$. □

Theorem 19 (Limit identification)**.** *$L = G$.*

Proof. Split the error at a parameter K:

$$\hat{P}_N - G\hat{Q}_N = \sum_{k=0}^{K} f(k)\varepsilon_k\, B(N,k) + \sum_{k=K+1}^{N} f(k)\varepsilon_k\, B(N,k),$$

where $\varepsilon_k = g(k)/f(k) - G$.

First sum: For fixed K, each $B(N,k)$ is a polynomial of degree $2k$ in N, so $\left|\sum_{k\le K}\right| \le C_K N^{2K}$. Dividing by $\hat{Q}_N \sim c\,\xi_+^N$ gives $C_K N^{2K}/\hat{Q}_N \to 0$.

Second sum: By Theorem 17, $f(k) > 0$ for all k. Since $B(N,k) > 0$,

$$\left|\sum_{k>K} f(k)\varepsilon_k B(N,k)\right| \le \max_{k>K} |\varepsilon_k| \cdot \sum_{k>K} f(k)B(N,k) \le \max_{k>K} |\varepsilon_k| \cdot \hat{Q}_N.$$

By Theorem 18, $\max_{k>K} |\varepsilon_k| \to 0$ as $K \to \infty$. Taking $N \to \infty$ then $K \to \infty$:

$$\limsup_{N\to\infty} \left|\frac{\hat{P}_N}{\hat{Q}_N} - G\right| \le \max_{k>K} |\varepsilon_k| \;\to\; 0. \qquad \square$$

15 Enhanced numerical verification

- Birkhoff eigenvector $v_+ = (1, r_{21}, r_{31})$ computed at 500-digit precision ($N_{\max} = 350$).
- $L = G$ confirmed to 500 digits.
- Delannoy decomposition: $f(k) > 0$ verified exactly for $k = 0, \ldots, 400$ (401 exact rational terms).
- $g(k)/f(k) = G$ to 40+ digits for $k \ge 56$.
- Error ratio $\varepsilon_{k+1}/\varepsilon_k \to -1/8$ (12 significant figures at $k = 118$).
- k-recurrence operators L_f, L_g verified exactly on all 401 terms; Poincaré polynomial $(\xi - 1)^2(\xi + 1/8)^6$ confirmed algebraically.

16 Status

Problem 2.5 is solved.

The limit $L = G$ (Catalan's constant) is proved unconditionally by Theorem 19, which relies on:

1. Delannoy basis decomposition (Theorem 15);
2. positivity $f(k) > 0$ for all k (Theorem 17, via exact verification for $k \le 400$ and Poincaré asymptotics for $k > 400$);
3. convergence $g(k)/f(k) \to G$ at geometric rate $-1/8$ (Theorem 18, via the common annihilator $\mathrm{LCLM}(L_f, L_g)$).

Additional established results: scalar N-recurrence (order 3, degree pattern $(7, 14, 21, 28)$, Poincaré roots -16, $-16(17 \pm 12\sqrt{2})$); k-recurrence (order 8, degree 42, Poincaré roots 1 and $-1/8$); Brafman identity with $k(\rho) = 1$; inverse Delannoy closed form.

Note: the irrationality of Catalan's constant remains open (this is not required by Problem 2.5).

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